

Models

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Naive Bayesian Logistic Regression Model

Priors

$$\beta_j \sim \mathcal{N}(0, 3), \quad \text{for } j \in \{0, 1, 2, 3, 4, 43\}.$$

Linear Predictor

$$\begin{aligned} \eta_i = & \beta_0 + \beta_1 \cdot \text{sex}_{i,\text{male}} + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{brain_vol}_i + \beta_4 \cdot \text{educ}_i \\ & + \beta_{34} \cdot (\text{brain_vol}_i \times \text{educ}_i). \end{aligned}$$

Link Function

$$\theta_i = \text{logistic}(\eta_i)$$

Likelihood

$$\text{CDR_binary}_i \mid \theta_i \sim \text{Bernoulli}(\theta_i).$$